Many Daughters – Pre-Analysis Plan

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This document pre-specifies the analysis for studying the impact of having daughters on attitudes, preferences and behaviours.

1 Variables

1.1 Outcome Variables

- **Outcome Variable for Hypothesis 1: Gender Equality (Measured by Attitudes)** To measure respondents' attitudes towards gender equality, we combined variables:

    1. `plh0303`, which asks whether men should be involved in housework. Responses of "strongly agree" or "somewhat agree" are coded as 1, and "somewhat disagree" or "strongly disagree" are coded as 0.
    2. `plh0301`, which states that women should focus more on family than career. Here, we reverse the coding: responses of "somewhat disagree" or "strongly disagree" are coded as 1 (egalitarian), and agreement is coded as 0 (traditional).
    3. `plh0309`, which states that working mothers can provide the same emotional warmth as non-working mothers. Agreement with this statement is coded as 1, disagreement as 0.

    We recode all three items to set invalid responses (e.g., "not included", "no answer") as missing values. We then constructed a composite index of gender-egalitarian attitudes, `gender_equality_attitude_index`. We use the `swindex` command, which creates a covariance-weighted index of the standardized and harmonized binary variables. Negative values indicate less progressive attitudes relative to the sample average. Since these variables were collected only in the 2012 wave, the analysis is limited to a cross-sectional design and does not make use of the panel dimension of the dataset.

- **Outcome Variable for Hypothesis 2: Left-Leaning Political Preferences**. To measure respondents' political orientation, we use the variable `plh0004`, which captures

self-placement on a left–right political spectrum ranging from 0 ("far left") to 10 ("far right"). We first recode invalid responses to missing, then reverse the scale (−`plh0004`) so that higher values indicate more left-wing orientation. Finally, we standardize this flipped score to mean=0, SD=1, yielding `political_score`. This z-score facilitates comparability across respondents and over time. Available SOEP waves: 2005, 2009, 2014, and 2019.

- **Outcome Variable for Hypothesis 3: Gender Equality Within the Household (Measured by Behaviors and Outcomes).** To capture gender equality in the household, we construct the variable `diff_hours`, based on the original variable `pgtatzeit`, which reports the number of hours worked. Invalid responses (e.g., "does not apply" or "no answer") are recoded as missing values. `diff_hours` measures the directional difference in hours worked between the female and male partner within a couple (i.e., `hours_female - hours_male`). A negative value indicates that the woman works fewer hours than the man, while a positive value indicates the opposite. To construct this variable, the dataset is reshaped to the couple level, so that each observation corresponds to a couple rather than an individual.

- **Outcome Variable for Hypothesis 4: Gender Equality (Measured by Attitudes).** The outcome variable used to test Hypothesis 4 is the same as that used for Hypothesis 1, capturing respondents' gender attitudes. For a detailed description of the variable and its construction, see the explanation provided above.

1.2 Explanatory Variables

- **Main Explanatory Variable:** `d_daughter` — A binary, time-varying indicator equal to 1 if the respondent has had at least one daughter born by the survey year, 0 if no daughters have been born, and missing if the respondent has no children. This variable is constructed as follows:

  1. Recode child gender variables (`kidsex*`) into `kidgender*`, where 1 = daughter and 0 = son.
  2. For each child, identify whether a daughter was born in or before the survey year.
  3. Define `d_daughter` as the maximum across these indicators — it equals 1 once a daughter is observed and remains 1 thereafter (absorbing state). Missing if the respondent reports not having children.

- **Other Controls:**

  1. `d_son` —A binary, time-varying indicator equal to 1 if the respondent has had at least one daughter born by the survey year, 0 if no daughters have been born, and missing if the respondent has no children. This variable is constructed in the same way as `d_daughter`, but based on the number of sons.

2. `kids_variant` — A time-varying variable equal to the total number of children born to the respondent up to and including the current survey year. It increases over time as additional children are born and remains constant thereafter if no new children are added.
3. `marital_status` — A categorical variable derived from the original variable `pgfamstd`, indicating the respondent's marital status. The categories are coded as follows:
    1. Married
    2. Married, but separated
    3. Single
    4. Divorced
    5. Widowed / Surviving partner from a registered same-sex partnership
    6. Married, spouse abroad
    7. Registered same-sex partnership, living together
    8. Registered same-sex partnership, living apart

    Responses coded as "not valid" or "no answer" are recoded as missing values.
4. `age` — A continuous variable representing the respondent's age, calculated as the difference between the survey year (`syear`) and the respondent's year of birth (`gebjahr`).
5. `age_sq` — A continuous variable capturing the squared value of age, to account for potential non-linear age effects in the analysis.
6. `female` — A binary variable indicating the respondent's gender. It is coded as 1 for female and 0 for male.
7. `years_education` — A continuous variable measuring the total years of education, derived from the original variable `pgbilzeit`. Responses coded as `-1` (no answer) or `-2` (not applicable) are recoded as missing.
8. `employment_status` — A categorical variable indicating the respondent's current employment status, based on the original variable `pgemplst`. Values equal to -1 (invalid or non-response) are recoded as missing.
9. `sum_kids` — A continuous variable indicating the total number of children reported by the respondent.
10. `syear` — The survey year in which the interview took place. It is included in the analysis to account for time fixed effects, capturing common shocks or trends across survey waves.
11. `corigin` — A categorical variable indicating the respondent's country of origin.

2 Inclusion/Exclusion Criteria

Sample for Hypothesis 1,2,4 : We restrict the sample to private households interviewed in Germany between 2000 and 2020.

Sample for Hypothesis 3 : For the analysis of Hypothesis 3, the data are reshaped so that each observation corresponds to a couple rather than an individual. In this case, we further restrict the sample to couples in which both partners report having the same number of children. Homosexuals couples have been excluded.

3 Exact Analysis

3.1 Hypothesis 1

To test Hypothesis 1, we estimate the following equation:

$$Y_i = \alpha + \beta_1 \text{Daughter}_i + \beta_2 \mathbf{X}_i + \varepsilon_i, \tag{1}$$

where Y_i denotes the gender equality attitude of respondent i. The coefficient β_1 captures the relationship between having a daughter and having more or less gender-progressive views. The vector X_i includes age, age squared, gender, whether the respondent has a son, the total number of children, years of education, employment status and country of origin. Robust standard errors are used to account for heteroskedasticity. Since the attitudinal variables are only available in the 2012 wave, the analysis is limited to a cross-sectional design and does not exploit the panel structure of the SOEP dataset.

3.2 Hypothesis 2

To test Hypothesis 2, we estimate the following equation:

$$Y_{it} = \alpha + \beta_1 \text{Daughter}_{it} + \beta_2 \mathbf{X}_{it} + \theta_t + \chi_i + \varepsilon_{it} \tag{2}$$

where Y_{it} denotes the left-leaning political preferences of respondent i, at survey year t. The coefficient β_1 captures the relationship between having a daughter and having more or less left-leaning political preferences. The vector X_{it} includes time-varying characteristics such as whether the respondent has a son, the total number of children, years of education, and employment status. Conditioning on the total number of respondents' children is of first-order importance because it allows us to separate the effect of parenting an additional child from the impact of parenting an additional daughter. χ_i refers to individual fixed effects, to take into account any time-invariant individual characteristics. θ_t represents survey year fixed effects. Standard errors are clustered at the individual level.

3.3 Hypothesis 3

To test Hypothesis 3, we estimate the following equation:

$$Y_{ct} = \alpha + \beta_1 \text{Daughter}_{ct} + \beta_2 \mathbf{X}_{ct} + \chi_c + \theta_t + \varepsilon_{ct} \tag{3}$$

where Y_{ct} denotes gender equality in behaviors (for example, hours worked) of the couple c, in the survey year t. The coefficient β_1 captures the relationship between having a daughter and being more or less gender equal within the couple. The vector X_{it} includes time-varying characteristics such as whether the respondent has a son, the total number of children, years of education, and employment status. Conditioning on the total number of repondents' children is of first-order importance because it allows us to separate the effect of parenting an additional child from the impact of parenting an additional daughter. χ_c refers to couples' fixed effects, to take into account any time-invariant characteristics. θ_t represents fixed effects for the survey year. Standard errors are clustered at the individual level.

3.4 Hypothesis 4

To test Hypothesis 4, we estimate the following equation:

$$Y_i = \alpha + \beta_1 \text{Daughter}_i + \beta_2 \text{Female}_i + \beta_3 \text{Female}_i \times \text{Daughter}_i + \beta_2 \mathbf{X}_i + \varepsilon_i \tag{4}$$

where Y_i denotes the attitude towards gender equality of the respondent i. The coefficient β_1 captures the relationship between having a daughter and having more or less gender-progressive views, conditional on being male. The coefficient β_2 measures the difference in gender attitudes between the female and male respondents. The interaction term β_3 captures whether the effect of having a daughter on gender attitudes differs by the respondent's gender. The vector X_i includes age, age squared, gender, whether the respondent has a son, the total number of children, years of education, employment status and country of origin. Robust standard errors are used to account for heteroskedasticity. Since the attitudinal variables are only available in the 2012 wave, the analysis is limited to a cross-sectional design and does not exploit the panel structure of the SOEP dataset.

4 Interpretation of the Results

4.1 Hypothesis 1

Table 1 reports a positive coefficient on the main independent variable—having a daughter—in the regression examining the relationship between daughters' presence in the household and support for gender equality. This indicates that the presence of a daughter is associated with more progressive gender attitudes within the household, in line with our initial hypothesis.

Table 1: Test Hypothesis 1

Coefficient	SE	t-statistic	Residual DF	Model DF	p-value	Num obs	Num participants	Interpretation
0.1884	0.1579	1.1932	291	29	0.2338	336	336	in line with hypothesis

4.2 Hypothesis 2

Table 2 shows that the regression coefficient of the main independent variable—having a daughter— is positive, indicating that the presence of a daughter in the household is associated to more left-leaning political preferences. This is in line with our initial hypothesis.

Table 2: Hypothesis 2

Coefficient	SE	t-statistic	Residual DF	Model DF	p-value	Num obs	Num participants	Interpretation
0.148	0.226	0.653	972	17	0.514	2581	973	in line with hypothesis

4.3 Hypothesis 3

Table 3 reports a negative coefficient on the main independent variable—having a daughter—when examining household gender equality, measured through behavioral outcomes, namely the division of work hours between parents. This result suggests that having daughters does not promote greater gender equality within the household, contrary to our initial hypothesis.

Table 3: Hypothesis 3

Coefficient	SE	t-statistic	Residual DF	Model DF	p-value	Num obs	Num couples	Interpretation
-1.427	3.317	-0.430	408	14	0.667	2505	409	not in line with hypothesis

4.4 Hypothesis 4

Table 4 shows that having daughters leads to a larger increase in the support of gender equality for women than for men, as the interaction term indicates. The sign of the coefficient (positive) is therefore not in line with the hypothesis that having daughters leads to a larger increase in the support of gender equality for men than for women.

Table 4: Hypothesis 4

Variable	Coefficient	SE	t-stat	Residual DF	Model DF	p-value	Num obs	Num participants	Interpretation
d_daughter	0.011	0.220	0.050	290	30	0.960	336	336	
d_daughter_female	0.302	0.244	1.242	290	30	0.215	336	336	not in line with hypothesis